

ARTUR IASENOVETS

Chongqing, China • archee.busy@gmail.com • [LinkedIn](#) • [ORCID](#) • [Portfolio](#)

Applied Cryptography Researcher focused on fully homomorphic encryption (FHE), private information retrieval (PIR), and privacy-preserving systems. M.S. student in Network & Information Security at Chongqing University of Posts and Telecommunications. Hands-on experience implementing BGV-based PIR in Hyperledger Fabric using Lattigo and CKKS-based encrypted aggregation for federated reinforcement learning using TenSEAL, PyTorch, and NumPy. Research background spans FHE, PIR, searchable encryption, cyber threat intelligence, and privacy-preserving computation, with emphasis on working prototypes, parameter selection, and empirical evaluation.

Research Interests: Fully Homomorphic Encryption, Private Information Retrieval, Privacy-Preserving Machine Learning, Secure Aggregation, Blockchain Privacy, Cyber Threat Intelligence

Education

- **Chongqing University of Posts and Telecommunications** Chongqing, China
M.S. Network & Information Security Supervisor: [Prof. Tang Fei](#)
GPA: 3.9/4.0
Sept. 2024 – Present
- **Ukhta State Technical University** Russia
B.S. Information Systems and Technologies
GPA: 4.5/5.0 (3.6/4.0)
March 2022 – June 2024
- **St. Petersburg State University of Telecommunications named after Prof. M.A. Bonch-Bruевич** Russia
B.S. Information Systems and Technologies
Sept. 2020 – March 2022

Publications & Manuscripts

- Bringing Private Reads to Hyperledger Fabric via Private Information Retrieval** Nov. 2025
lasenovets A., Tang F., Zhu H., Wang P., Liu L. – arXiv preprint (submitted to IEEE) [arXiv:2511.02656](#)
Proposed and implemented a BGV homomorphic encryption-based PIR mechanism natively in Hyperledger Fabric chain-code, enabling private world-state reads without modifying Fabric's core architecture; achieved 113 ms end-to-end latency and ~2 MB peer traffic per query across a multi-channel prototype.
- EVMK-SSE: Efficient and Verifiable Multi-Keyword Symmetric Searchable Encryption** Dec. 2025
Zhu H., Tang F., Qin W., Shan J., Wang P., lasenovets A. – J. King Saud Univ. Comput. Inf. Sci. [DOI](#)
Proposed EVMK-SSE, replacing DH-OPRF with a lightweight OT-based client-independent relaxed OPRF in a multi-keyword SSE scheme, reducing computational overhead while maintaining strong privacy guarantees for outsourced encrypted cloud databases.
- MFCTIIF: Multi-Feed Cyber Threat Intelligence Integration Framework** Jan. 2026
lasenovets A.V., Tang F. – International Research Journal, No. 1 (163) [DOI](#)
Proposed a modular CTI framework aggregating heterogeneous feeds (MalwareBazaar, VirusTotal, APT ETDA), performing metadata enrichment, synonym resolution, and automated threat-level classification; evaluated on 100 malware samples with ~15–17s end-to-end latency.
- Adaptive Privacy-Aware Federated Reinforcement Learning with Trust-Weighted Secure Aggregation** 2026
Manuscript in preparation – second author
Contributing CKKS-based encrypted aggregation using TenSEAL with chunked model-update encoding for privacy-preserving federated reinforcement learning; implementation uses PyTorch and NumPy for local learning/evaluation and TenSEAL for encrypted aggregation.

Selected Research Projects

BGV-Based Private Information Retrieval for Hyperledger Fabric

- Implemented a single-server HE-PIR prototype using BGV over packed CTI records, with client-side encrypted selection vectors and chaincode-side ciphertext-plaintext evaluation.
- Designed fixed-window record packing, parameter/capacity checks, and client-side decoding for privacy-preserving reads from Fabric world state.
- Evaluated end-to-end latency, peer-side evaluation time, network traffic, storage overhead, and parameter/capacity constraints across multiple Fabric channels.
- **Technologies:** Go, Lattigo, Hyperledger Fabric, Docker, BGV, PIR, RLWE-based HE.

CKKS-Based Encrypted Aggregation for Federated Reinforcement Learning

- Implemented CKKS-encrypted model-update aggregation for federated reinforcement learning using TenSEAL, PyTorch, and NumPy.
- Used chunking to encode model-update vectors into encrypted CKKS payloads for aggregation.
- Compared plaintext, differential privacy, trust-weighted, HE, and combined HE+DAPI execution paths for network-defense learning.
- **Technologies:** Python, TenSEAL, PyTorch, NumPy, CKKS, federated learning, reinforcement learning.

Technical Skills

Cryptography: Fully Homomorphic Encryption, BGV, CKKS, PIR, searchable encryption, OPRF/OT-based constructions, differential privacy, secure aggregation, RLWE/LWE-based security assumptions

FHE Libraries: Lattigo, Microsoft SEAL, TenSEAL, OpenFHE; familiar with TFHE and Concrete documentation

Programming: Python, Go, TypeScript, JavaScript, SQL

ML / Data: PyTorch, NumPy, federated learning, reinforcement learning, CIC-IDS2017-style intrusion detection workflows

Systems / Security: Hyperledger Fabric, Docker, GitHub Actions, S3/MinIO, Airflow, YARA, Chainsaw, MITRE ATT&CK

Language Skills: [English C1](#), [Chinese HSK3](#), Russian native

Research Experience

Graduate Student Researcher

Chongqing University of Posts and Telecommunications – Applied Cryptography Group

Chongqing, China

Sept. 2024 – Present

- Designed and implemented BGV-based private information retrieval for Hyperledger Fabric using Lattigo, including packed plaintext encoding, encrypted query generation, ciphertext-plaintext evaluation, and client-side decoding.
- Implemented CKKS-based encrypted aggregation for federated reinforcement learning using TenSEAL, PyTorch, and NumPy, including chunked model-update encoding.
- Evaluated privacy-preserving prototypes using latency, communication volume, storage overhead, ciphertext expansion, and parameter/capacity constraints.
- Contributed to research papers on FHE-based PIR, searchable encryption, cyber threat intelligence, and privacy-preserving federated learning.
- Participating in weekly research seminars, paper reviews, and collaborative research discussions within the group.

Professional Experience

Security Intern

Remote

Positive Technologies — Leading Eastern European cybersecurity company

Aug. 2024 – Feb. 2025

- Developed Python utilities for the MaxPatrol EDR product team, including regex-based Windows path/network configuration parsing, unit-test coverage, and a Windows Prefetch-based software allowlist/blocklist checker.
- Engineered a Python malware-processing pipeline for the Threat Intelligence team, fetching samples from VX-Underground, storing artifacts in S3/MinIO, scanning with YARA rules, and packaging the workflow as an Airflow DAG.
- Performed SOC analysis on CVE-2020-5902 and CVE-2023-38831 attack chains, including EVTX log forensics with Chainsaw, PowerShell deobfuscation, VBA macro analysis with oletools, IoC extraction, and MITRE ATT&CK mapping.

Software Engineer

Remote

academai.ru — AI-powered EdTech startup under Sberbank's Accelerator program

Aug. 2023 – Aug. 2024

- Built a Flask REST API backend where each endpoint encapsulates a structured system prompt, orchestrating LLM calls via Langchain to generate individual course components on demand.
- Developed a responsive frontend with Next.js, TypeScript, Prisma, and Auth.js, ensuring seamless authentication and consistent UI across the platform.
- Reduced deployment friction by building a CI/CD pipeline via GitHub Actions with Docker containerization.

Honors & Achievements

Chinese Government Scholarship for Graduate Studies

Sept. 2024 – Aug. 2027

Awarded a full scholarship by the Chinese government for the entire duration of the M.S. program at CQUPT.

5 Science Conferences

Apr. 2022 – Apr. 2023

Won 3 consecutive first places in paper evaluations across various scientific conferences while at USTU.

Kaspersky Cyberimmune Development Hackathon 2.0

Dec. 2023

Participated in Kaspersky's Cyberimmunity Hackathon in Fall 2023, placing 11th out of 100+ teams.

MIPT & Sber Phystech GigaChat Challenge Hackathon

Dec. 2023

Developed an LLM application for generating personalized educational courses using Sberbank's GigaChat and Kandinsky APIs within a 48-hour sprint; later presented at the regional final demo day.